Error Analysis of a DSP Multirate System (Quantizer → Decimator → Interpolator → Polyphase Filter)

ECA0905-DIGITALSIGNAL PROCESSING

GUIDED BY
DR.MARY REEJA.Y

BY
PAVAN KUMAR S
192312420

OBJECTIVE

- To understand the architecture and operation of a DSP multirate processing system.
- To analyze the function of each stage: quantizer, decimator, interpolator, and polyphase FIR filter.
- To identify implementation errors that cause distortion, such as quantization noise, aliasing, imaging, ringing, amplitude distortion, and delay mismatch.
- To investigate the effect of these errors on both the time-domain and frequency-domain characteristics of the output signal.

PROBLEM STATEMENT

- **Input Signal → Quantizer → Decimator → Interpolator → Polyphase FIR Filter → Output**

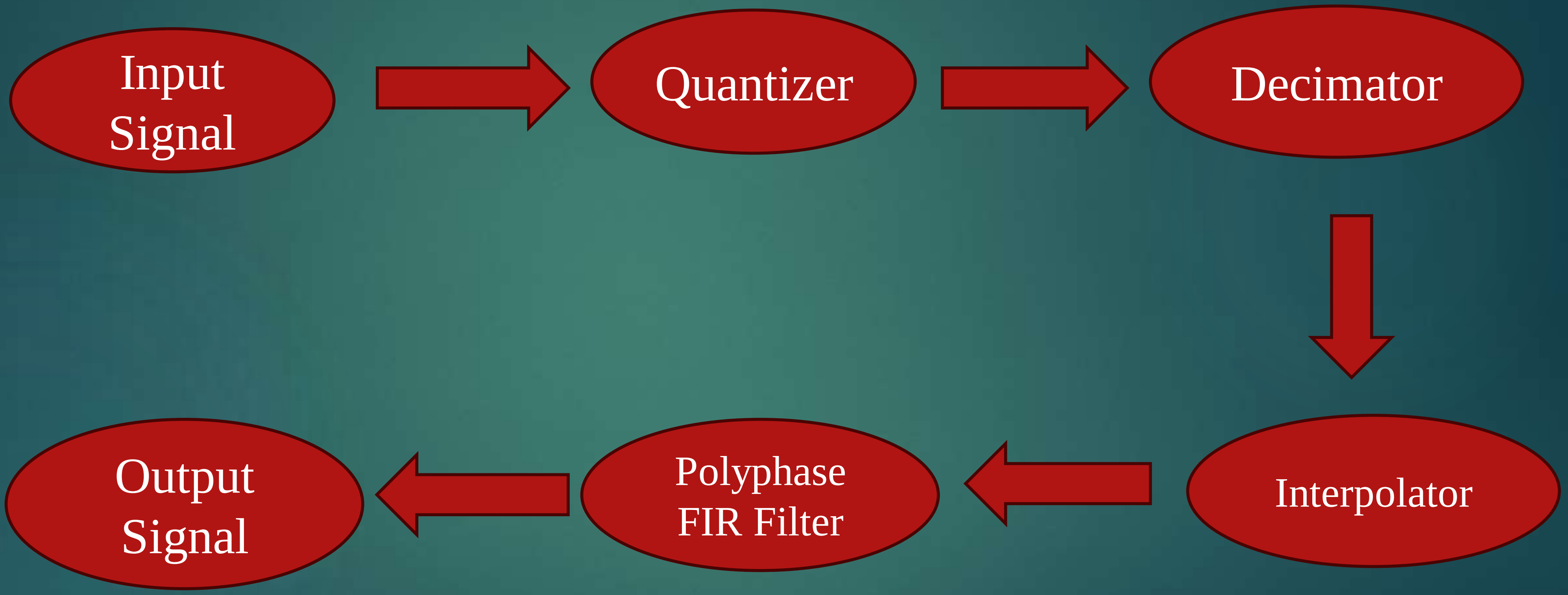
The output signal exhibits the following problems:

- Aliasing
- Oscillations (Ringing)
- Amplitude distortion
- Delay mismatch

TASK

- Analyze each processing stage.
- Identify implementation errors.
- Explain the reason for each error.
- Suggest suitable corrections with justification.

DSP MULTIRATE SYSTEM



ERROR 1 – QUANTIZATION ERROR

OBSERVATION
Output contains noise.
Signal quality decreases.

CAUSE
Low quantization resolution.
Few quantization levels.

EFFECT
Quantization noise.
Lower Signal-to-Noise Ratio (SNR).

CORRECTION
Increase quantization bits.
Example:
- 4-bit → 16 levels
- 8-bit → 256 levels
- 12-bit → 4096 levels

JUSTIFICATION
Higher resolution reduces quantization error and improves signal accuracy.

TASK
Analyze each processing stage.
Identify implementation errors.

ERROR 2 – DELAY MISMATCH

OBSERVATION
Output signal is shifted in time.

CAUSE
Group delay introduced by the FIR filter.

$$\text{Delay} = \frac{N-1}{2}$$

EFFECT
Input and output are not time-aligned.

CORRECTION
Compensate for the filter delay.

JUSTIFICATION
Produces proper synchronization between input and output.

ERROR 3 – AMPLITUDE DISTORTION

OBSERVATION
Output amplitude decreases.

CAUSE
Filter gain not normalized.

EFFECT
Incorrect signal magnitude.

CORRECTION
Normalize FIR coefficients:

$$\sum h\,(n) = 1$$

JUSTIFICATION
Maintains unity gain and preserves the original signal amplitude.

ERROR 4 – OSCILLATIONS (RINGING)

OBSERVATION
Output oscillates near sharp transitions.

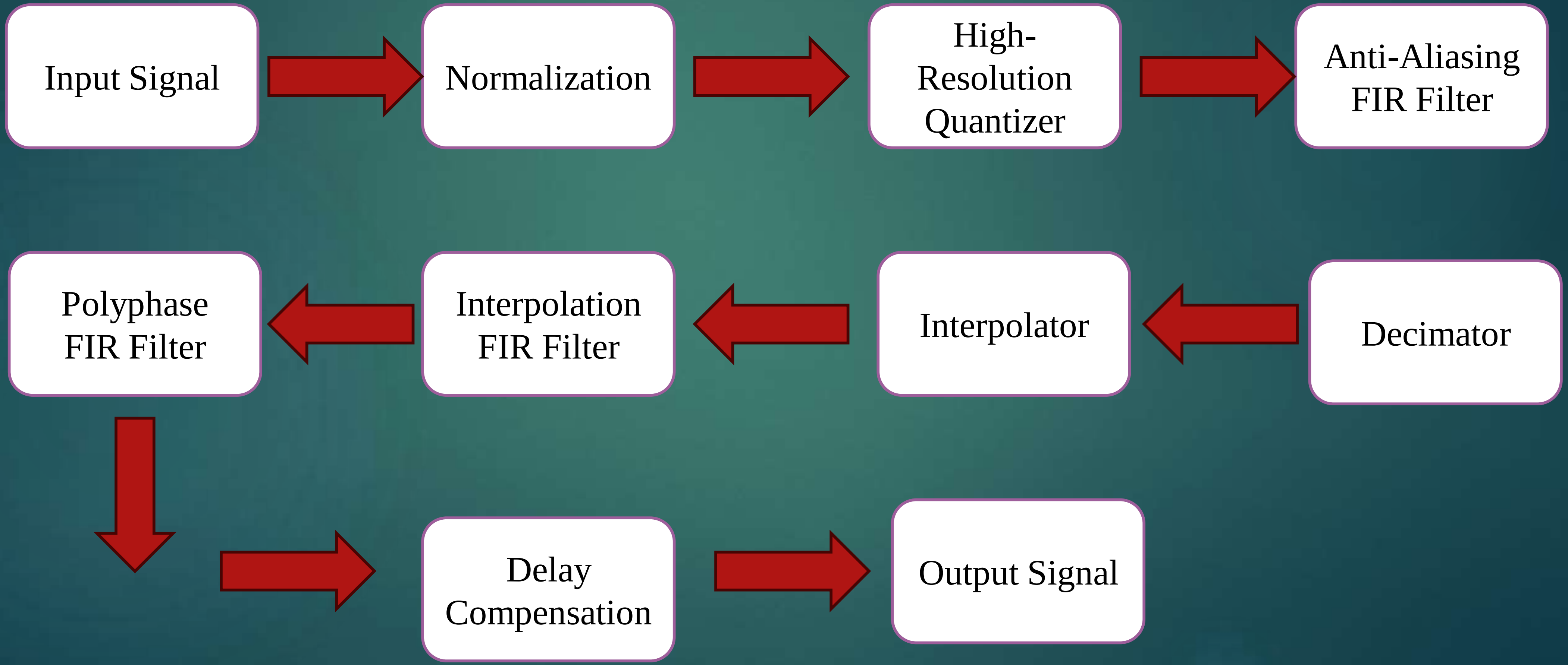
CAUSE
Poor FIR filter design.
Low filter order.
Inappropriate window.

Effect
Ringing
Overshoot
Reduced signal quality

CORRECTION
Increase filter order.
Use Hamming, Blackman, or Kaiser window.
Optimize filter coefficients.

JUSTIFICATION
Improves stopband attenuation and reduces oscillations.

CORRECTED DSP PROCESSING CHAIN



ADVANTAGES OF THE CORRECTED SYSTEM

1)Reduced quantization noise.
2)Eliminates aliasing.
3)Removes imaging components.
4)Reduces ringing effects.
5)Preserves signal amplitude.
6)Provides correct time alignment.
7)Improves overall Signal-to-Noise Ratio (SNR).

CONCLUSION

The distorted output in the DSP multirate system is caused by several implementation errors, including insufficient quantization resolution, missing anti-aliasing filtering before decimation, absence of interpolation filtering after upsampling, poor FIR filter design, unnormalized filter coefficients, and uncompensated filter delay. By increasing quantization resolution, adding appropriate low-pass filters, designing an optimized polyphase FIR filter, normalizing filter gain, and compensating for delay, the system achieves an alias-free, low-distortion, and accurately reconstructed output signal suitable for efficient multirate digital signal processing.


THANK YOU!